

Business Statement

Business Problem

A leading automotive dealership operating across multiple states handles hundreds of thousands of used vehicle transactions each year. While the company has extensive sales data, it lacks a centralized analytical framework to answer critical business questions related to pricing, inventory management, market performance, and regional sales trends.

Without data-driven insights, decision-makers face challenges such as:

- Identifying which manufacturers and vehicle models generate the highest revenue.
- Understanding whether vehicles are consistently sold above or below their estimated market value (MMR).
- Evaluating the impact of vehicle age, mileage, transmission type, and condition on selling prices.
- Determining which states and regions contribute the most to overall revenue.
- Recognizing seasonal sales patterns and demand fluctuations throughout the year.
- Measuring dealership profitability across different vehicle segments.
- Optimizing inventory by focusing on high-performing brands and vehicle categories.

The absence of these insights can result in inefficient pricing strategies, slower inventory turnover, reduced profitability, and missed market opportunities.

Step 1: Cleaning data in Python

Step 2: Drawing business insights from SQL queries

1. Total Revenue

```
SELECT  
ROUND(SUM(sellingprice),2) AS Total_Revenue  
FROM vehicle_sales;
```

	Total_Revenue
▶	7507207954

2. Total Vehicles Sold

```
SELECT  
COUNT(*) AS Vehicles_Sold  
FROM vehicle_sales;
```

	Vehicles_Sold
▶	548400

3. Average Selling Price

```
SELECT
ROUND(AVG(sellingprice),2) AS Avg_Selling_Price
FROM vehicle_sales;
```

	Avg_Selling_Price
▶	13689.29

4. Average Market Value (MMR)

```
SELECT
ROUND(AVG(mmr),2) AS Avg_Market_Value
FROM vehicle_sales;
```

	Avg_Market_Value
▶	13848.83

5. Average Vehicle Age

```
SELECT
ROUND(AVG(Vehicle_Age),2) AS Avg_Vehicle_Age
FROM vehicle_sales;
```

	Avg_Vehicle_Age
▶	4.79

6. Revenue by Manufacturer

```
SELECT
```

```

make,
COUNT(*) AS Vehicles_Sold,
ROUND(SUM(sellingprice),2) AS Revenue,
ROUND(AVG(sellingprice),2) AS Avg_Selling_Price
FROM vehicle_sales
GROUP BY make
ORDER BY Revenue DESC;

```

Result Grid					Filter Rows:		Export:	Wrap Cell Content:
	make	Vehicles_Sold	Revenue	Avg_Selling_Price				
▶	Ford	93996	1362577082	14496.12				
	Chevrolet	60587	723369949	11939.36				
	Nissan	54017	633703805	11731.56				
	Toyota	39966	489089925	12237.65				
	BMW	20707	440383013	21267.35				
	Mercedes-Benz	17141	370083906	21590.57				
	Dodge	30953	345561569	11164.07				
	Infiniti	15305	312489190	20417.46				
	Honda	27351	298273884	10905.41				
	Lexus	11980	242163555	20213.99				
	Hyundai	21831	240212386	11003.27				
	Jeep	15483	231513328	14952.74				

7. Top 10 Selling Models

```

SELECT
make,
model,
COUNT(*) AS Vehicles_Sold
FROM vehicle_sales
GROUP BY make, model
ORDER BY Vehicles_Sold DESC
LIMIT 10;

```

	make	model	Vehicles_Sold
▶	Nissan	Altima	19349
	Ford	F-150	14479
	Ford	Fusion	12945
	Toyota	Camry	12549
	Ford	Escape	11893
	Ford	Focus	10399
	Honda	Accord	9155
	BMW	3 Series	8204
	Chevrolet	Impala	7957
	Dodge	Grand Caravan	7941

8. Revenue by Body Type

```

SELECT
body,
COUNT(*) AS Vehicles,
ROUND(SUM(sellingprice),2) AS Revenue,
ROUND(AVG(sellingprice),2) AS Avg_Price
FROM vehicle_sales
GROUP BY body
ORDER BY Revenue DESC;

```

Result Grid   Filter Rows:				
	body	Vehicles	Revenue	Avg_Price
▶	Sedan	241234	2821131845	11694.59
	SUV	143844	2318177919	16115.92
	Crew Cab	16394	354914507	21649.05
	Minivan	25529	295340141	11568.81
	Coupe	17752	282750644	15927.82
	Hatchback	26237	263569910	10045.73
	SuperCrew	9033	199614784	22098.39
	Convertible	10476	186896701	17840.46
	Wagon	16128	163290866	10124.68
	G Sedan	7417	147918806	19943.21
	SuperCab	5311	67863591	12777.93
	Van	4528	62897335	13890.75

9. Automatic vs Manual

```

SELECT
transmission,

```

```

COUNT(*) AS Vehicles,
ROUND(AVG(sellingprice),2) AS Avg_Price
FROM vehicle_sales
GROUP BY transmission;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
transmission	Vehicles	Avg_Price	
automatic	467823	13614.84	
Unknown	63584	14907.75	
manual	16993	11179.79	

10. Vehicle Condition Performance

```

SELECT
Vehicle_Condition,
COUNT(*) AS Vehicles,
ROUND(AVG(sellingprice),2) AS Avg_Selling_Price
FROM vehicle_sales
GROUP BY Vehicle_Condition
ORDER BY Avg_Selling_Price DESC;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Vehicle_Condition	Vehicles	Avg_Selling_Price	
Excellent	158440	19412.64	
Good	170942	13470.32	
Poor	110173	10098.4	10098.4
Fair	108845	9336.72	

11. Market Performance Distribution

```

SELECT
Market_Status,
COUNT(*) AS Vehicles
FROM vehicle_sales
GROUP BY Market_Status;

```

Market_Status	Vehicles
Sold Near Market	347854
Sold Below Market	111154
Sold Above Market	89392

12. Average Premium Percentage

```
SELECT  
ROUND(AVG(Premium_Percentage),2) AS Avg_Premium  
FROM vehicle_sales;
```

	Avg_Premium
▶	-0.71

13. Premium Percentage by Manufacturer

```
SELECT  
make,  
ROUND(AVG(Premium_Percentage),2) AS Avg_Premium,  
COUNT(*) AS Vehicles  
FROM vehicle_sales  
GROUP BY make  
HAVING COUNT(*) > 100  
ORDER BY Avg_Premium DESC;
```

	make ▼	Avg_Premium	Vehicles
	Volvo	NULL	3788
	Volkswagen	NULL	12579
	Toyota	NULL	39966
	Suzuki	NULL	1078
	Subaru	NULL	5103
	smart	NULL	396
	Scion	NULL	1687
	Saturn	NULL	2841
	Saab	NULL	484
▶	Ram	NULL	4574
	Porsche	NULL	1402
	pontiac	NULL	4524

14. Highest Price Difference

```
SELECT  
make,
```

```

model,
Price_Difference
FROM vehicle_sales
ORDER BY Price_Difference DESC
LIMIT 20;

```

	make	model	Price_Difference
►	Toyota	Corolla	53650
	Nissan	Versa	57150
	Mercedes-Benz	Sprinter	52100
	Mercedes-Benz	S-Class	28000
	Mazda	Tribute	26500
	GMC	Savana	30475
	Ford	Escape	207200
	Ford	Escape	99950
	Ford	Fusion	33800
	Ford	Mustang	31800
	Ford	F-250 Su...	31000
	Chevrolet	Suburban	39700

15. Price Difference by Vehicle Condition

```

SELECT
Vehicle_Condition,
ROUND(AVG(Price_Difference),2) AS Avg_Difference
FROM vehicle_sales
GROUP BY Vehicle_Condition;

```

	Vehicle_Condition	Avg_Difference
►	Poor	-790.11
	Excellent	371.24
	Good	-11.33
	Fair	-526.66

16. Vehicle Age Analysis

```

SELECT
Vehicle_Age,
COUNT(*) AS Vehicles,
ROUND(AVG(sellingprice),2) AS Avg_Price
FROM vehicle_sales
GROUP BY Vehicle_Age

```

ORDER BY Vehicle_Age;

	Vehicle_Age	Vehicles	Avg_Price
▶	0	14312	23966.41
	1	82090	19807.12
	2	98915	17391.2
	3	98915	16585.8
	4	45871	15644.74
	5	25542	12310.35
	6	21709	10275.71
	7	31297	9370.62
	8	29219	7752.23
	9	24838	5907.13
	10	19830	4801.68
	11	16040	4168.73

17. Mileage Band Analysis

```
SELECT  
Mileage_Band,  
COUNT(*) AS Vehicles,  
ROUND(AVG(sellingprice),2) AS Avg_Price  
FROM vehicle_sales  
GROUP BY Mileage_Band;
```

	Mileage_Band	Vehicles	Avg_Price
▶	Very Low	117536	20838.88
	Low	150934	17471.74
	Moderate	148301	11809.98
	Very High	44454	3270.3
	High	87175	6010.9

18. Price Range Distribution

```
SELECT  
Price_Range,  
COUNT(*) AS Vehicles,  
ROUND(SUM(sellingprice),2) AS Revenue  
FROM vehicle_sales  
GROUP BY Price_Range;
```

	Price_Range	Vehicles	Revenue
►	Mid-Range	101224	2645504726
	Luxury	1515	119111492
	Economy	229193	3288794154
	Budget	108797	845972494
	Premium	7383	343998095
	Entry	100288	263826993

19. Revenue by State

```
SELECT
state,
COUNT(*) AS Vehicles,
ROUND(SUM(sellingprice),2) AS Revenue
FROM vehicle_sales
GROUP BY state
ORDER BY Revenue DESC;
```

	state	Vehicles	Revenue
►	fl	81480	1138819073
	ca	71658	1041929651
	pa	53154	852504673
	tx	45195	600920561
	ga	34056	442391061
	nj	27331	372424743
	tn	20691	353182908
	il	23188	344406801
	oh	21209	304988884
	mo	15798	229973961
	mi	15287	229043176
	nv	12453	188692288

20. Average Selling Price by State

```
SELECT
state,
ROUND(AVG(sellingprice),2) AS Avg_Price
FROM vehicle_sales
GROUP BY state
ORDER BY Avg_Price DESC;
```

	state	Avg_Price
▶	on	18232.23
	tn	17069.4
	pa	16038.39
	co	15985.26
	nv	15152.36
	mi	14982.87
	il	14852.8
	mo	14557.16
	ca	14540.31
	oh	14380.16
	wa	14077.44
	wi	14054.82

21. Market Status by State

```

SELECT
state,
Market_Status,
COUNT(*) AS Vehicles
FROM vehicle_sales
GROUP BY state, Market_Status;

```

	state	Market_Status	Vehicles
▶	ca	Sold Near Market	41683
	ca	Sold Below Market	13120
	ca	Sold Above Market	16855
	tx	Sold Near Market	27804
	pa	Sold Near Market	33386
	mn	Sold Below Market	1412
	az	Sold Near Market	5792
	wi	Sold Below Market	2215
	tn	Sold Near Market	12333
	md	Sold Near Market	7436
	fl	Sold Below Market	15839
	ne	Sold Below Market	546

22. Monthly Sales Trend

```

SELECT
Sale_Year,
Sale_Month_Num,
Sale_Month,
COUNT(*) AS Vehicles,
ROUND(SUM(sellingprice),2) AS Revenue
FROM vehicle_sales
GROUP BY Sale_Year, Sale_Month_Num, Sale_Month
ORDER BY Sale_Year, Sale_Month_Num;

```

	Sale_Year	Sale_Month_Num	Sale_Month	Vehicles	Revenue
►	2014	1	January	206	3204525
	2014	2	February	1	10500
	2014	12	December	53414	602977655
	2015	1	January	138506	1849422432
	2015	2	February	159089	2184133454
	2015	3	March	45259	613735647
	2015	4	April	1393	14385843
	2015	5	May	51915	747368044
	2015	6	June	97379	1470793340
	2015	7	July	1238	21176514

23. Quarterly Revenue

```

SELECT
Sale_Year,
Sale_Quarter,
ROUND(SUM(sellingprice),2) AS Revenue
FROM vehicle_sales
GROUP BY Sale_Year, Sale_Quarter
ORDER BY Sale_Year, Sale_Quarter;

```

	Sale_Year	Sale_Quarter	Revenue
►	2014	1	3215025
	2014	4	602977655
	2015	1	4647291533
	2015	2	2232547227
	2015	3	21176514

24. Sales by Day of Week

```

SELECT
Sale_Day,
COUNT(*) AS Vehicles,
ROUND(SUM(sellingprice),2) AS Revenue
FROM vehicle_sales
GROUP BY Sale_Day;

```

	Sale_Day	Vehicles	Revenue
▶	Tuesday	177199	2408531330
	Wednesday	143606	2041536481
	Thursday	76725	1060472708
	Monday	128184	1712289868
	Friday	11179	163899567
	Sunday	11507	120478000

25. Top 10 Sellers

```

SELECT
seller,
COUNT(*) AS Vehicles_Sold,
ROUND(SUM(sellingprice),2) AS Revenue
FROM vehicle_sales
GROUP BY seller
ORDER BY Revenue DESC
LIMIT 10;

```

	seller	Vehicles_Sold	Revenue
▶	ford motor credit company llc	19147	340821525
	nissan-infiniti lt	19693	272104560
	the hertz corporation	18277	250703067
	nissan infiniti lt	9962	216158321
	avis corporation	12533	203343030
	financial services remarketing (lease)	5732	163584300
	mercedes-benz financial services	5657	152754200
	enterprise veh exchange/rental	6836	126069587
	santander consumer	15040	119993640
	tdaf remarketing	7089	114659125

26. Brand Profitability Analysis

```

SELECT
make,
ROUND(AVG(sellingprice - mmr),2) AS Avg_Profitability
FROM vehicle_sales
GROUP BY make
ORDER BY Avg_Profitability DESC;

```

	make	Avg_Profitability
►	airstream	41500
	Aston Martin	1252
	Lamborghini	1125
	mazda tk	1125
	Lotus	500
	gmc truck	429.55
	HUMMER	252.62
	hyundai tk	225
	landrover	197.22
	chev truck	75
	Suzuki	15.19
	Isuzu	-41.66

27. Brand Market Share

```

SELECT
make,
COUNT(*) AS Vehicles_Sold,
ROUND(
(COUNT(*) * 100.0) / (SELECT COUNT(*) FROM vehicle_sales),
2
) AS Market_Share_Percentage
FROM vehicle_sales
GROUP BY make
ORDER BY Market_Share_Percentage DESC;

```

	make	Vehicles_Sold	Market_Share_Percentage
►	Ford	93996	17.14
	Chevrolet	60587	11.05
	Nissan	54017	9.85
	Toyota	39966	7.29
	Dodge	30953	5.64
	Honda	27351	4.99
	Hyundai	21831	3.98
	BMW	20707	3.78
	Kia	18082	3.30
	Chrysler	17483	3.19
	Mercede...	17141	3.13
	Jeep	15483	2.82

28. Mileage Impact on Selling Price

```

SELECT
  Mileage_Band,
  COUNT(*) AS Vehicles,
  ROUND(AVG(odometer), 0) AS Avg_Mileage,
  ROUND(AVG(sellingprice), 2) AS Avg_Selling_Price,
  ROUND(AVG(mmr), 2) AS Avg_Market_Value
FROM vehicle_sales
GROUP BY Mileage_Band
ORDER BY Avg_Mileage;

```

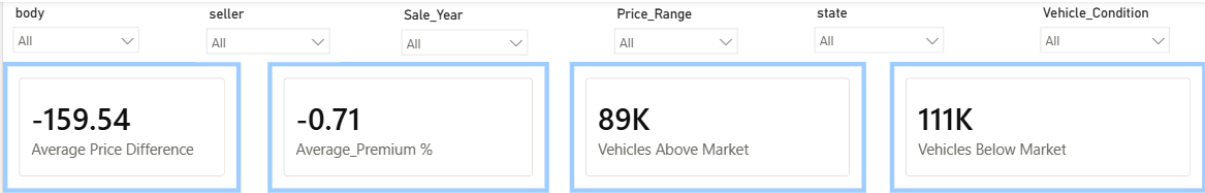
	Mileage_Band	Vehicles	Avg_Mileage	Avg_Selling_Price	Avg_Market_Value
►	Very Low	117536	15387	20838.88	21108.9
	Low	150934	36431	17471.74	17577.65
	Moderate	148301	72516	11809.98	11886.56
	High	87175	121897	6010.9	6276.37
	Very High	44454	187814	3270.3	3388.8

29. Body Type Performance

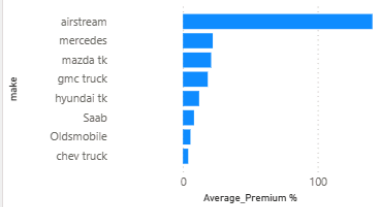
```

SELECT
  body,
  COUNT(*) AS Vehicles_Sold,
  ROUND(SUM(sellingprice), 2) AS Revenue,
  ROUND(AVG(sellingprice), 2) AS Avg_Selling_Price
FROM vehicle_sales
GROUP BY body
ORDER BY Revenue DESC;

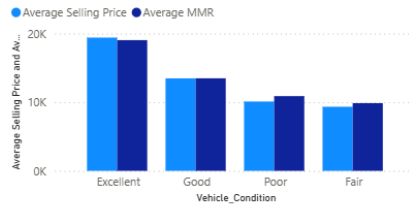
```

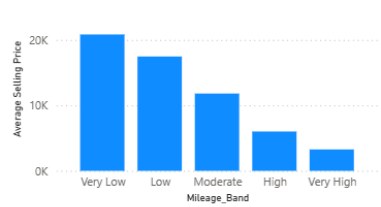
Average Premium (%) by Manufacturer



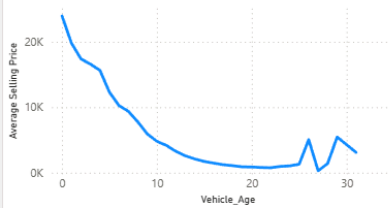
Selling Price vs Market Value by Vehicle Condition



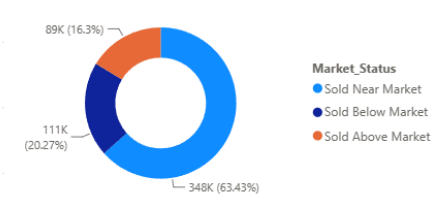
Average Selling Price by Mileage Band



Average Selling Price by Vehicle Age



Vehicle Distribution by Market Status



Revenue Contribution by Body Type

